
Concept Questions: Chapter 7

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ME-3640 – Heat Transfer

2026-Spring



Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts fo the remainder of this chapter.

1 Drag Force and Heat Transfer in External Flow

- 7-1C** What is the difference between the upstream velocity and the free-stream velocity? For what types of flow are these two velocities equal to each other?
- 7-2C** What is drag? What causes it? Why do we usually try to minimize it?
- 7-3C** What is lift? What causes it? Does wall shear contribute to the lift?
- 7-4C** During flow over a given body, the drag force, the upstream velocity, and the fluid density are measured. Explain how you would determine the drag coefficient. What area would you use in calculations?
- 7-5C** Define frontal area of a body subjected to external flow. When is it appropriate to use the frontal area in drag and lift calculations?
- 7-6C** What is the difference between skin friction drag and pressure drag? Which is usually more significant for slender bodies such as airfoils?
- 7-7C** What is the difference between streamlined and blunt bodies? Is a tennis ball a streamlined or blunt body?
- 7-8C** What is the effect of streamlining on (a) friction drag and (b) pressure drag? Does the total drag acting on a body necessarily decrease as a result of streamlining? Explain.
- 7-9C** What is the effect of surface roughness on the friction drag coefficient in laminar and turbulent flows?
- 7-10C** What is flow separation? What causes it? What is the effect of flow separation on the drag coefficient?

2 Flow over Flat Plates

- 7-11C** What does the friction coefficient represent in flow over a flat plate? How is it related to the drag force acting on the plate?
- 7-12C** Consider laminar flow over a flat plate. Will the friction coefficient change with distance from the leading edge? How about the heat transfer coefficient?
- 7-13C** How are the average friction and heat transfer coefficients determined in flow over a flat plate?

3 Flow across Cylinders and Spheres

- 7-60C** In flow over blunt bodies such as a cylinder, how does the pressure drag differ from the friction drag?

- 7-61C** In flow over cylinders, why does the drag coefficient suddenly drop when the flow becomes turbulent? Isn't turbulence supposed to increase the drag coefficient instead of decreasing it?
- 7-62C** Why is flow separation in flow over cylinders delayed in turbulent flow?
- 7-63C** Consider laminar flow of air across a hot circular cylinder. At what point on the cylinder will the heat transfer be highest? What would your answer be if the flow were turbulent?

4 Flow across Tube Banks

- 7-99C** In flow across tube banks, how does the heat transfer coefficient vary with the row number in the flow direction? How does it vary with in the transverse direction for a given row number?
- 7-100C** In flow across tube banks, why is the Reynolds number based on the maximum velocity instead of the uniform approach velocity?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
Homework is completed with appropriate effort, neatness, etc.	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.